

09. Working with the Pipeline and Pipeline-Specialized Commands

Aim

To use the PowerShell pipeline with Where-Object, Sort-Object, Measure-Object, and Select-Object.

Where-Object

```
Get-Service | Where-Object {$_.Status -eq 'Running'}  
Get-Process | Where-Object {$_.WorkingSet64 -gt 100MB}
```

Sort-Object

```
Get-Process | Sort-Object WorkingSet64 -Descending  
Get-Service | Sort-Object DisplayName
```

Select-Object

```
Get-Process | Select-Object Name, Id, CPU  
Get-Process | Select-Object -First 5
```

Measure-Object

```
Get-Process | Measure-Object  
Get-ChildItem | Measure-Object  
Get-ChildItem *.txt | Measure-Object Length -Sum
```

Combined Pipeline Example

```
Get-Process |  
Where-Object {$_.WorkingSet64 -gt 100MB} |  
Sort-Object WorkingSet64 -Descending |  
Select-Object -First 10 Name, Id, WorkingSet64
```

Exercise

- Display only running services.
- Display the five processes using the most memory.
- Count the number of running processes.
- Sort files in a folder by size.
- Create a pipeline that filters, sorts, and selects process information.